Bandbreite der Programmiersprachen

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99 Bottles of Beer

99 bottles of beer on the wall, 99 bottles of beer. Take one down and pass it around, 98 bottles of beer on the wall.
98 bottles of beer on the wall, 98 bottles of beer. Take one down and pass it around, 97 bottles of beer on the wall.
[. . .]
2 bottles of beer on the wall, 2 bottles of beer. Take one down and pass it around, 1 bottle of beer on the wall.
1 bottle of beer on the wall, 1 bottle of beer. Take one down and pass it around, no more bottles of beer on the wall.
No more bottles of beer on the wall, no more bottles of beer. Go to the store and buy some more, 99 bottles of beer on the wall.

Quelle: nach "Lyrics of the song 99 Bottles of Beer" on 99-bottles-of-beer.net

Imperativ, Hardwarenah: C

```c
#define MAXBEER (99)
void chug(int beers);
main() {
   register beers;
   for(beers = MAXBEER; beers; chug(beers--))  puts("");
   puts("\nTime to buy more beer!\n");
}
void chug(register beers) {
   char howmany[8], *s;
   s = beers != 1 ? "s" : "";
   printf("%d bottle%s of beer on the wall,\n", beers, s);
   printf("%d bottle%s of beeeeer . . . ,\n", beers, s);
   printf("Take one down, pass it around,\n");
   if(--beers) sprintf(howmany, "%d", beers); else strcpy(howmany, "No more");
   s = beers != 1 ? "s" : "";
   printf("%s bottle%s of beer on the wall.\n", howmany, s);
}
```

Quelle: “Language C” by Bill Wein on 99-bottles-of-beer.net

Imperativ, Objektorientiert: Java

```java
class bottles {
    public static void main(String args[]) {
        String s = "s";
        for (int beers=99; beers>-1;) {
            System.out.print(beers + " bottle" + s + " of beer on the wall, ");
            System.out.println(beers + " bottle" + s + " of beer, ");
            if (beers==0) {
                System.out.print("Go to the store, buy some more, ");
                System.out.println("99 bottles of beer on the wall.\n");
                System.exit(0);
            } else
                System.out.print("Take one down, pass it around, ");
            s = (--beers == 1)?"":"s";
            System.out.println(beers + " bottle" + s + " of beer on the wall.\n");
        }
    }
}
```

Quelle: “Language Java” by Sean Russell on 99-bottles-of-beer.net

Logisch: Prolog

```
bottles :-
    bottles(99).

bottles(1) :-
    write('1 bottle of beer on the wall, 1 bottle of beer,'), nl,
    write('Take one down, and pass it around,'), nl,
    write('Now they are all gone.'), nl,!.
bottles(X) :-
    write(X), write(' bottles of beer on the wall,'), nl,
    write(X), write(' bottles of beer,'), nl,
    write('Take one down and pass it around,'), nl,
    NX is X - 1,
    write(NX), write(' bottles of beer on the wall.'), nl, nl,
    bottles(NX).
```

Quelle: "Language Prolog" by M@ on 99-bottles-of-beer.net

Funktional: Haskell

```haskell
bottles 0 = "no more bottles"
bottles 1 = "1 bottle"
bottles n = show n ++ " bottles"

verse 0   = "No more bottles of beer on the wall, no more bottles of beer.\n"
         ++ "Go to the store and buy some more, 99 bottles of beer on the wall."

verse n   = bottles n ++ " of beer on the wall, " ++ bottles n ++ " of beer.\n"
         ++ "Take one down and pass it around, " ++ bottles (n-1)
                                                  ++ " of beer on the wall.\n"

main      = mapM (putStrLn . verse) [99,98..0]
```

Quelle: “Language Haskell” by Iavor on 99-bottles-of-beer.net

Brainfuck

Quelle: Screenshot of “Language Brainfuck” by Michal Wojciech Tarnowski on 99-bottles-of-beer.net

Programmiersprache Lox

```
fun fib(x) {
    if (x == 0) {
        return 0;
    } else {
        if (x == 1) {
            return 1;
        } else {
            fib(x - 1) + fib(x - 2);
        }
    }
}

var wuppie = fib;
wuppie(4);
```

Wrap-Up

- Compiler übersetzen formalen Text in ein anderes Format
- Berücksichtigung von unterschiedlichen
  - Sprachkonzepten (Programmierparadigmen)
  - Typ-Systemen
  - Speicherverwaltungsstrategien
  - Abarbeitungsstrategien

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Exceptions

Citations of “99 Bottles of Beer” lyrics and programming language examples (license(s) unknown):

- “Lyrics of the song 99 Bottles of Beer” on 99-bottles-of-beer.net (Author unknown, Website hosted by Oliver Schade, Gregor Scheithauer, and Stefan Scheler)
- “Language C” by Bill Wein on 99-bottles-of-beer.net
- “Language Java” by Sean Russell on 99-bottles-of-beer.net
- “Language Prolog” by M@ on 99-bottles-of-beer.net
- “Language Haskell” by Iavor on 99-bottles-of-beer.net
- Screenshot of “Language Brainfuck” by Michal Wojciech Tarnowski on 99-bottles-of-beer.net (file `screenshot_brainfuck_99bottles.png`)